

RBM, Jastrow and Mean-Field Wavefunctions for the Transverse-Field Ising Chain: A Small Controlled VMC Study with SGD and Stochastic Reconfiguration

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Abstract—We compare restricted-Boltzmann-machine (RBM) wavefunctions at hidden densities $\alpha \in \{1, 2, 4\}$ with a mean-field product state and a Jastrow ansatz on the periodic 1D transverse-field Ising model with $N = 8$ –12 spins, using variational Monte Carlo with plain SGD or stochastic reconfiguration (SR) and exact diagonalisation as the reference. Ten systems, 13 (size, field) tasks, 5 seeds and two starts (symmetric and symmetry-broken initial fields) give 1300 main runs; per seed the lower-energy start is kept and evaluated by exact enumeration. The $\alpha=2$ RBM with SR has a lower mean energy error than mean-field and Jastrow with SR in all 13 tasks. Against Jastrow the margin is a factor of 8.55 to 71 for fields Γ between 1 and 1.5, where Jastrow is at the best energy that a sampling-free L-BFGS optimisation finds for it, but only 1.47 to 2.37 for $\Gamma \leq 0.5$, where in three of four tasks the sampling-free Jastrow optimum is below the RBM’s Monte Carlo result, so that margin reflects optimisation rather than the ansatz. From the symmetric start alone, Jastrow stays above mean-field at $\Gamma \leq 0.5$ and all 28 diverged runs (RBM with SR) occur. For the $\alpha=2$ RBM, SR has a lower mean error than SGD in all 13 tasks, by a factor of 2.07 at $\Gamma=0.25$ and of 68 to 496 for $\Gamma \geq 1$. At the critical field the $\alpha=4$ RBM with SR is lower than the $\alpha=2$ RBM on every seed. Two of five hypotheses were refuted: mean-field energy error does not peak at the critical field, and RBM-SR error there grows with N . In the ordered phase the lower-energy states are often symmetry-broken, with fidelity near 0.5. This is a CPU-scale study of one model with five seeds and reimplemented baselines; it supports no broader claim.

Index Terms—neural quantum states, restricted Boltzmann machine, variational Monte Carlo, stochastic reconfiguration, transverse-field Ising model

I. Introduction

Neural quantum states (NQS) represent a many-body wavefunction by a neural network and optimise it variationally. Carleo and Troyer introduced the restricted

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Boltzmann machine (RBM) as a variational ansatz, optimised with Monte Carlo sampling and stochastic reconfiguration (SR), and tested it on the transverse-field Ising and Heisenberg models [1]. Since then the idea has spread to autoregressive [2], [3] and convolutional [4] wavefunctions, to better optimisers for large networks [5], and to standard software [6] and benchmark collections [7].

At the smallest scales, where exact diagonalisation (ED) is cheap, the questions are different from those at scale and are rarely isolated in one table: how much of an RBM’s accuracy comes from the ansatz and how much from the optimiser, how it compares with the two simplest classical variational families under the same optimiser and sample budget, and whether an accurate energy implies an accurate order parameter across a quantum phase transition. The one-dimensional transverse-field Ising model (TFIM) [8] is a good test bed: it has a critical point at $\Gamma = J$, a nodeless ground state (so real positive amplitudes suffice), and for $N \leq 12$ the full Hilbert space has at most 4096 states.

We study this on a small numpy testbed. The ansaetze are a mean-field (product) state, a two-body Jastrow state [9], and RBMs with $\alpha = 1, 2, 4$ hidden units per spin; the optimisers are plain SGD and SR [10]. We pre-registered five hypotheses (Sec. III), ran 1300 main runs (two starts), and report which were supported. Contributions are modest and empirical: (i) an optimiser-by-ansatz grid with exactly evaluated errors in energy, magnetisation and fidelity; (ii) a start ablation and a sampling-free reference optimisation that separate ansatz error from optimisation error; (iii) an account of failures (divergence of SR, stuck symmetric starts, symmetry-broken low-energy states) and protocol revisions. We make no claim of a new method.

II. Related Work

Neural quantum states. The RBM wavefunction and its SR optimisation were introduced in [1]. Representational

aspects of Boltzmann machines and deeper variants were studied in [11], and RBMs combined with pair-product states were used for strongly correlated fermions in [12]. Autoregressive models allow exact sampling [2], [3], and symmetric convolutional networks were applied to the frustrated J_1 - J_2 model [4]. Our RBM is the original shallow form, with real weights and no symmetries.

Optimisation. SR [10] preconditiones the energy gradient with the covariance matrix of the log-derivatives of the wavefunction; the resulting update has the form of a natural-gradient step [13]. MinSR [5] makes this tractable for networks with many parameters; our networks have at most a few hundred parameters, so dense SR is used.

Classical variational baselines. Product (mean-field) states and two-body Jastrow wavefunctions [9] are two simple classical variational families; we use them as baselines. Both are reimplemented here and trained with the same two optimisers and the same budget as the RBMs.

Benchmarks and tools. NetKet [6] provides production implementations of RBM, SR and ED; the variational benchmark collection of [7] reports energies and variances for larger systems. We used neither, so our absolute numbers are not comparable with theirs. The TFIM chain is exactly solvable by a mapping to free fermions [8]; we use ED as the reference instead because it gives the ground-state vector, from which the fidelity and $\langle |M_z| \rangle$ follow directly.

III. Method

Model. $H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - \Gamma \sum_i \sigma_i^x$ with periodic boundaries and $J = 1$. Configurations are $s \in \{\pm 1\}^N$; wavefunctions are real and positive, $\psi_\theta(s) = \exp(\log \psi_\theta(s))$.

Ansätze. Mean-field (MF): $\log \psi = \sum_i a_i s_i$ (N parameters). Jastrow: $\log \psi = \sum_i a_i s_i + \sum_{i < j} J_{ij} s_i s_j$ ($N + N(N-1)/2$ parameters, no translation symmetry imposed). RBM with $H = \alpha N$ hidden units:

$$\log \psi = \sum_i a_i s_i + \sum_{j=1}^H \log 2 \cosh \left(b_j + \sum_i W_{ji} s_i \right), \quad (1)$$

with $N + H + HN$ real parameters.

Variational Monte Carlo. The local energy is $E_{\text{loc}}(s) = -J \sum_i s_i s_{i+1} - \Gamma \sum_i \psi(s^{(i)})/\psi(s)$, where $s^{(i)}$ flips spin i . Samples $s \sim |\psi|^2$ come from 256 persistent Metropolis chains (90% single-spin flips, 10% global flips). With $O_k = \partial_{\theta_k} \log \psi$, the energy gradient is

$$g_k = 2(\langle E_{\text{loc}} O_k \rangle - \langle E_{\text{loc}} \rangle \langle O_k \rangle). \quad (2)$$

SGD: $\theta \leftarrow \theta - \eta g$. SR: $\theta \leftarrow \theta - \eta(S + \epsilon I)^{-1} g$, with $S_{kl} = \langle O_k O_l \rangle - \langle O_k \rangle \langle O_l \rangle$ and diagonal shift ϵ . Both use the same samples; no momentum, Adam, schedule or step clipping.

Two starts. All parameters are drawn from $\mathcal{N}(0, 0.01^2)$. Start S (symmetric) uses them as drawn; start B (symmetry-broken) adds 0.5 to every field a_i , which biases the initial state towards one of the two ordered configurations. Every (system, task, seed) is optimised

from both starts with the same rate and budget, and the selected state is the one with the lower variational energy E ; a diverged start loses to a finite one. The selection needs no reference energy; we apply it to the exactly evaluated E .

Evaluation. For each trained state we sum over all 2^N configurations to obtain exactly $E = \langle H \rangle$, $\langle |M_z| \rangle$ with $M_z = N^{-1} \sum_i s_i$, $\langle \sigma^x \rangle = N^{-1} \sum_i \langle \sigma_i^x \rangle$, and the fidelity $F = |\langle \psi_0 | \psi \rangle|^2$ with the ED ground state ψ_0 (scipy eigsh). The metrics are $\epsilon_E = (E - E_0)/|E_0|$, ϵ_{M_z} (absolute difference to ED; likewise for $\langle \sigma^x \rangle$) and $1 - F$. ($\langle M_z \rangle = 0$ in the symmetric finite-size ground state, hence $|M_z|$.)

Sampling-free reference. To separate what an ansatz can represent from what VMC reaches, we also minimise the enumerated energy $E(\theta) = \sum_s p_\theta(s) E_{\text{loc}}(s)$, $p_\theta \propto \psi_\theta^2$, with L-BFGS (at most 1000 iterations) and the exact gradient, i.e. Eq. (2) with averages over p_θ instead of samples. The lower of the two starts is an upper bound on the best ϵ_E of the ansatz.

Hypotheses (registered in proposal.md before the main runs). H1: RBM-SR ($\alpha=2$) has lower ϵ_E than MF and Jastrow (with SR) on every task. H2: SR beats SGD for the RBM ($\alpha=2$) on every task. H3: with SR, ϵ_E decreases with α ($1 > 2 > 4$) at $\Gamma=1$ for each N . H4: MF energy and $|M_z|$ errors peak within 0.25 of $\Gamma=1$ (for $N=10$). H5: RBM-SR error at $\Gamma=1$ does not grow from $N=8$ to 12. The registered tests are the per-task ordering of means over the 5 seeds and a paired comparison over seeds. The two-start selection was added after registration (Sec. IV); the tests are applied to the selected states, with per-start results in Table VI.

IV. Experimental Setup

Tasks. $N = 10$ with $\Gamma \in \{0.25, 0.5, 0.75, 1, 1.25, 1.5, 2\}$ and $N \in \{8, 12\}$ with $\Gamma \in \{0.5, 1, 1.5\}$: 13 tasks, $J = 1$, written N/Γ below. Systems. {MF, Jastrow, RBM $\alpha = 1, 2, 4$ } $\times$ {SGD, SR}: 10 systems, all reimplemented in method/run.py; the method of record is RBM $\alpha=2$ with SR, the rest are baselines. Seeds. Main runs use seeds 0–4 and both starts ($13 \times 10 \times 5 \times 2 = 1300$ runs); the seed fixes the initial parameters and the sampler. Tuning uses disjoint seeds 100–102.

Budget. Every system and start gets 300 iterations with 256 samples per iteration (10N burn-in steps, N Metropolis steps between iterations) and SR shift $\epsilon = 10^{-2}$. The learning rate is the only tuned quantity, per (ansatz, optimiser), from the grid $\{0.003, 0.01, 0.03, 0.1\}$ for SGD and $\{0.02, 0.05, 0.1\}$ for SR, by the lowest mean ϵ_E over 3 tasks (10/0.25, 10/1, 12/0.5) and 3 tuning seeds from start S, a diverged run counting as infinite error. The SGD grid has one more point, so tuning cost differs by optimiser (Table VIII). Selected rates: 0.1 for MF (both optimisers) and RBM $\alpha=2, 4$ with SR; 0.05 for RBM $\alpha=1$ with SR; 0.03 for Jastrow and all RBMs with SGD; 0.02 for Jastrow with SR. Rates were not re-tuned for start B.

Protocol history. Two earlier protocols (v1: rate tuned on one task and one seed; v2: 3 tasks, one seed) are superseded rows in the registry (Table VIII). An audit of the single-start results found two faults, both fixed here: Jastrow was stored as an $N \times N$ block with only the upper triangle used (it now has one parameter per pair; all Jastrow runs were repeated), and from start S the baselines were stuck in poor states at small fields. Start B and the two-start selection were therefore added for all ten systems, after the hypotheses were registered.

Reporting. Unless a start is named, table cells are medians over the 5 seeds of the selected state. A run is “diverged” if its parameters became non-finite. Hardware: shared CPU, 2 threads, numpy. Run counts and times are in Table VIII; the brief’s target of about 30 minutes for all experiments was exceeded. Code, logs and the run registry are in the repository; `experiments/make_tables.py` builds all tables and figures from the registry.

V. Results

H1 (RBM-SR $\alpha=2$ beats MF and Jastrow): supported by the registered tests; at small fields the margin over Jastrow is not an ansatz effect. The mean ϵ_E of RBM $\alpha=2$ SR is below that of MF-SR and Jastrow-SR in all 13 tasks, and it wins 65 of 65 paired seeds against MF-SR and 64 of 65 against Jastrow-SR (Table II). The paired test gives $p < 0.05$ in all 13 tasks against Jastrow-SR and in 12 against MF-SR ($p = 0.207$ at 10/0.5); with $n = 5$ and uncorrected tests these are indicative only. The ratio of means to Jastrow is 1.47 to 2.37 in the four $\Gamma \leq 0.5$ tasks, 1.59 at 10/2, 3.80 at 10/0.75, and 8.55 to 71 in the seven tasks with Γ between 1 and 1.5. The L-BFGS rows of Table I qualify these margins. For $\Gamma \geq 0.75$ the Jastrow-SR error is within about 30% of the best L-BFGS value of the Jastrow ansatz (at 10/1, 1.5×10^{-3} against 1.3×10^{-3}), and the RBM-SR median is below that value in all nine such tasks: better optimisation of this Jastrow form would not close the gap. For $\Gamma \leq 0.5$ the Jastrow-SR error is above its L-BFGS value, slightly at 8/0.5 and by more than two orders of magnitude at 10/0.25 (5.6×10^{-5} against 2.5×10^{-7}), and that L-BFGS value is below the RBM-SR median in three of the four tasks (not at 8/0.5). In those three tasks the ordering of the two VMC results reflects optimisation and sampling under this budget, not what the Jastrow ansatz can represent. The MF median is close to its L-BFGS value in every task (largest gap at 10/0.5, 1.3×10^{-3} against 1.0×10^{-3}): it is limited by the ansatz.

H2 (SR beats SGD, RBM $\alpha=2$): supported. The SGD-to-SR ratio of mean errors exceeds one in all 13 tasks (Table II): 2.07 at 10/0.25, 15 to 23 in the four tasks at $\Gamma=0.5$ and 0.75, and 68 to 496 in the eight tasks with $\Gamma \geq 1$; SR wins 65 of 65 paired seeds, with $p \leq 0.015$ in every task. The cost is robustness from start S, where SR diverged in 12 of 65 runs ($\alpha=2$) and 16 of 65 ($\alpha=4$); no SGD run and no run from start B diverged (Table VIII).

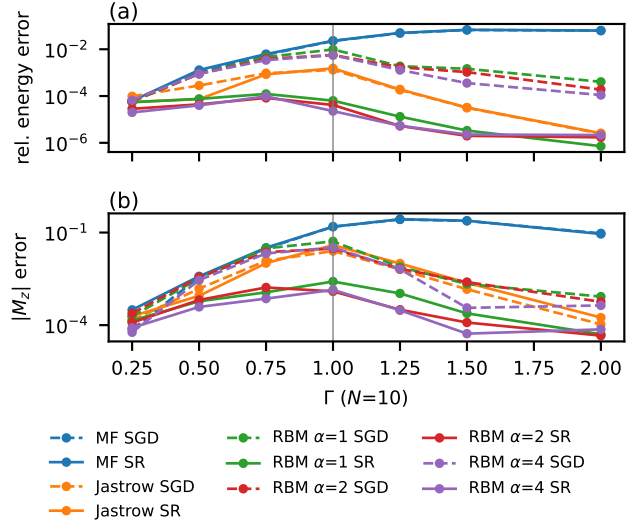


Fig. 1. (a) Median relative energy error and (b) median $|M_z|$ error of the selected states versus Γ at $N = 10$ (solid: SR, dashed: SGD). Vertical line: $\Gamma = J$.

For MF and Jastrow the optimisers differ far less, at most by a factor of about 5 (Jastrow at 12/0.5, Table I).

H3 (error decreases with α at $\Gamma=1$ with SR): supported, with overlapping seed ranges. Mean and median ϵ_E are ordered $\alpha=1 > 2 > 4$ for $N = 8, 10$ and 12 (Table III). Not every step exceeds the seed spread: at $N=12$ the lowest $\alpha=1$ seed (8.2×10^{-5}) is below the highest $\alpha=2$ seed (8.8×10^{-5}), and at $N=10$ the ranges of $\alpha=2$ and 4 overlap. So the method of record is not the best system at the critical field: RBM $\alpha=4$ SR is lower on 5 of 5 seeds at $\Gamma=1$ for each N , and over all tasks $\alpha=2$ wins 31 of 65 seeds against it (Table II). The ordering is not general: at 10/2 $\alpha=1$ has the lowest median (Table I). Nor is it a capacity limit: at 10/1 L-BFGS reaches 2.2×10^{-6} with $\alpha=2$, below every value in Table III, so a larger α may ease the stochastic optimisation rather than add needed capacity; we did not test this.

H4 (MF errors peak near $\Gamma=1$): refuted for energy. At $N=10$ the median MF energy error rises up to $\Gamma=1.5$ (6.8×10^{-2} with SR) and is slightly lower at $\Gamma=2$ (Table I, Fig. 1a): the maximum is more than 0.25 from the critical field. The L-BFGS reference has the same maximum, so this is a property of the product state, not of the optimiser. The MF $|M_z|$ error peaks at $\Gamma=1.25$ (Table IV), on the boundary of the registered window.

H5 (RBM-SR error does not grow with N): refuted. At $\Gamma=1$ the mean ϵ_E of RBM $\alpha=2$ SR is 1.7×10^{-5} , 4.0×10^{-5} and 7.9×10^{-5} for $N = 8, 10, 12$, and $\alpha=1$ and 4 grow likewise (Table III). With three sizes and a fixed budget we cannot separate ansatz, budget and sampler.

Magnetisation and fidelity: where low energy misleads. Magnetisation errors are far larger than energy errors: for RBM $\alpha=2$ SR at 10/1, $\epsilon_{M_z} = 1.2 \times 10^{-3}$ against

TABLE I

Relative energy error ϵ_E of the selected state (lower-energy start), median over 5 seeds; columns are N/Γ . MF: mean-field, Jas: Jastrow; P_{10} : number of parameters at $N=10$. Last three rows: sampling-free L-BFGS reference (lower of two starts, seed 0).

System	P_{10}	10/0.25	10/0.5	10/0.75	10/1	10/1.25	10/1.5	10/2	8/0.5	8/1	8/1.5	12/0.5	12/1	12/1.5
MF SGD	10	6.3e-5	1.0e-3	5.8e-3	2.3e-2	4.9e-2	6.7e-2	6.3e-2	1.1e-3	2.5e-2	6.7e-2	9.9e-4	2.1e-2	6.7e-2
MF SR	10	6.2e-5	1.3e-3	6.2e-3	2.3e-2	5.0e-2	6.8e-2	6.3e-2	1.1e-3	2.6e-2	6.7e-2	1.0e-3	2.2e-2	6.8e-2
Jas SGD	55	9.8e-5	2.8e-4	9.4e-4	1.3e-3	1.8e-4	3.1e-5	2.5e-6	2.9e-4	1.1e-3	3.6e-5	3.2e-4	1.4e-3	3.0e-5
Jas SR	55	5.6e-5	7.4e-5	8.4e-4	1.5e-3	1.9e-4	3.2e-5	2.6e-6	1.4e-4	1.2e-3	3.7e-5	6.0e-5	1.7e-3	3.0e-5
RBM $\alpha=1$ SGD	120	5.7e-5	8.9e-4	4.6e-3	9.8e-3	1.9e-3	1.5e-3	4.1e-4	9.2e-4	1.0e-2	5.0e-4	8.7e-4	7.7e-3	1.4e-3
RBM $\alpha=1$ SR	120	5.4e-5	7.5e-5	1.2e-4	6.3e-5	1.3e-5	3.3e-6	7.2e-7	6.2e-5	3.3e-5	2.4e-6	5.7e-5	9.4e-5	5.5e-6
RBM $\alpha=2$ SGD	230	6.0e-5	8.9e-4	3.9e-3	5.7e-3	1.7e-3	1.1e-3	1.9e-4	9.1e-4	9.3e-3	3.6e-4	9.1e-4	6.0e-3	9.1e-4
RBM $\alpha=2$ SR	230	2.8e-5	4.4e-5	8.3e-5	4.1e-5	5.3e-6	2.0e-6	1.7e-6	5.8e-5	1.7e-5	7.7e-7	3.7e-5	8.0e-5	3.1e-6
RBM $\alpha=4$ SGD	450	6.6e-5	8.9e-4	3.4e-3	5.6e-3	1.3e-3	3.6e-4	1.1e-4	9.2e-4	2.3e-3	3.1e-4	9.1e-4	2.1e-3	7.7e-4
RBM $\alpha=4$ SR	450	2.0e-5	4.0e-5	9.7e-5	2.3e-5	5.4e-6	2.3e-6	2.1e-6	5.2e-5	7.5e-6	1.0e-6	4.2e-5	4.6e-5	5.8e-6
MF L-BFGS	10	6.1e-5	1.0e-3	5.7e-3	2.2e-2	4.8e-2	6.6e-2	6.0e-2	1.1e-3	2.5e-2	6.6e-2	9.8e-4	2.1e-2	6.6e-2
Jas L-BFGS	55	2.5e-7	3.0e-5	8.0e-4	1.3e-3	1.7e-4	2.9e-5	2.2e-6	1.0e-4	1.1e-3	3.5e-5	1.7e-5	1.3e-3	2.7e-5
RBM $\alpha=2$ L-BFGS	230	6.9e-8	1.6e-5	3.8e-6	2.2e-6	9.0e-7	7.9e-8	1.7e-8	8.6e-5	3.2e-7	3.0e-8	3.5e-6	6.2e-6	9.7e-8

TABLE II

Registered tests on selected states, RBM $\alpha=2$ SR (R2-SR) against MF-SR, Jastrow-SR, RBM $\alpha=2$ SGD (R2-SGD) and RBM $\alpha=4$ SR (R4-SR): seeds (of 5) won by R2-SR; ratio of mean ϵ_E to that of R2-SR; two-sided paired t -test over n seeds, uncorrected.

N/Γ	seeds won by R2-SR against				mean ϵ_E ratio to R2-SR			paired t -test p			n
	MF-SR	Jas-SR	R2-SGD	R4-SR	MF-SR	Jas-SR	R2-SGD	MF-SR	Jas-SR	R2-SGD	
10/0.25	5/5	5/5	5/5	3/5	2.49	1.95	2.07	0.043	0.020	0.009	5
10/0.5	5/5	4/5	5/5	1/5	72	1.47	17	0.207	0.038	<0.001	5
10/0.75	5/5	5/5	5/5	3/5	30	3.80	17	<0.001	0.010	<0.001	5
10/1	5/5	5/5	5/5	0/5	586	38	139	<0.001	<0.001	0.010	5
10/1.25	5/5	5/5	5/5	2/5	9612	37	317	<0.001	<0.001	<0.001	5
10/1.5	5/5	5/5	5/5	4/5	33162	16	496	<0.001	<0.001	<0.001	5
10/2	5/5	5/5	5/5	4/5	38751	1.59	118	<0.001	0.006	<0.001	5
8/0.5	5/5	5/5	5/5	2/5	19	2.37	15	<0.001	<0.001	<0.001	5
8/1	5/5	5/5	5/5	0/5	1509	71	424	<0.001	<0.001	0.015	5
8/1.5	5/5	5/5	5/5	5/5	76682	43	407	<0.001	<0.001	<0.001	5
12/0.5	5/5	5/5	5/5	3/5	28	1.59	23	<0.001	0.004	<0.001	5
12/1	5/5	5/5	5/5	0/5	282	22	68	<0.001	<0.001	0.002	5
12/1.5	5/5	5/5	5/5	4/5	19277	8.55	255	<0.001	<0.001	<0.001	5
Total /65	65	64	65	31							

TABLE III

RBM with SR at $\Gamma=1$, selected states: ϵ_E over 5 seeds.

N	α	mean	median	min	max
8	1	3.3e-5	3.3e-5	2.8e-5	3.7e-5
8	2	1.7e-5	1.7e-5	1.5e-5	2.1e-5
8	4	8.8e-6	7.5e-6	6.6e-6	1.5e-5
10	1	6.4e-5	6.3e-5	5.7e-5	7.5e-5
10	2	4.0e-5	4.1e-5	3.1e-5	4.5e-5
10	4	2.6e-5	2.3e-5	2.1e-5	3.7e-5
12	1	1.1e-4	9.4e-5	8.2e-5	1.8e-4
12	2	7.9e-5	8.0e-5	6.9e-5	8.8e-5
12	4	5.1e-5	4.6e-5	4.4e-5	6.5e-5

TABLE IV

$|M_z|$ error of the selected state at $N=10$ versus Γ (median over 5 seeds); R2: RBM $\alpha=2$.

Γ	MF-SR	Jas-SR	R2-SGD	R2-SR
0.25	3.1e-4	1.9e-4	2.4e-4	1.2e-4
0.5	3.8e-3	8.9e-4	3.6e-3	6.5e-4
0.75	3.2e-2	1.0e-2	2.3e-2	1.7e-3
1	1.5e-1	3.9e-2	3.0e-2	1.2e-3
1.25	2.7e-1	9.9e-3	6.7e-3	3.2e-4
1.5	2.4e-1	2.4e-3	2.5e-3	1.2e-4
2	9.0e-2	1.8e-4	5.7e-4	4.6e-5

$\epsilon_E = 4.1 \times 10^{-5}$ (Tables IV, I). At 10/1 the MF-SR states have a median infidelity of 0.59 and the RBM $\alpha=2$ SGD states 0.26 (Table V). In the ordered phase the selected states fail in a different way: Jastrow-SR is selected from start B on all five seeds for $\Gamma \leq 0.75$ with $1 - F = 0.50$, and the RBMs at $\Gamma=0.5$ are selected from B on four of five seeds with the same median infidelity. These states have ϵ_E below 10^{-4} at $\Gamma=0.5$ but overlap with only half of the

symmetric finite- N ground state; they may approximate one of the two ordered configurations, whose energies are nearly degenerate at small Γ . Energy is a poor selection criterion in this regime, and $\langle |M_z| \rangle$ does not detect the difference.

VI. Ablations and Analysis

Start ablation. Table VI gives the two starts separately. Neither dominates. From start S, Jastrow stays far above

TABLE V

Infidelity $1 - F$ of the selected state at $N=10$ versus Γ (median over 5 seeds); in brackets, the number of seeds whose selected state comes from start B.

System	0.25	0.5	0.75	1
MF SR	5.0e-1 (3)	5.0e-1 (2)	5.2e-1 (3)	5.9e-1 (2)
Jas SR	5.0e-1 (5)	5.0e-1 (5)	5.0e-1 (5)	6.4e-3 (0)
RBM $\alpha=1$ SR	7.9e-3 (2)	5.0e-1 (4)	8.2e-4 (0)	1.5e-4 (1)
RBM $\alpha=2$ SR	5.4e-2 (2)	5.0e-1 (4)	2.9e-3 (1)	9.1e-5 (2)
RBM $\alpha=4$ SR	3.2e-2 (1)	5.0e-1 (4)	4.6e-4 (0)	5.6e-5 (2)
RBM $\alpha=2$ SGD	5.0e-1 (5)	5.0e-1 (4)	5.0e-1 (0)	2.6e-1 (0)

its start-B result for $\Gamma \leq 0.75$ (SR at 10/0.5: 3.0×10^{-3} against 7.4×10^{-5}) and above the selected MF error in all four $\Gamma \leq 0.5$ tasks, although the Jastrow family contains the product state; MF-SR at 12/0.5 has a median of 2.3×10^{-1} , a domain-wall state, against 1.0×10^{-3} from start B; and all 28 diverged runs are RBM-SR runs from start S. From start B, Jastrow is worse at $\Gamma=1$ (SR at 10/1: 4.1×10^{-3} against 1.5×10^{-3}) and the RBMs with SR are worse at 10/0.75 ($\alpha=2$: 7.6×10^{-4} against 8.1×10^{-5}). With start S alone, as registered, H1 also holds on medians, but with margins over Jastrow at small fields that are artefacts of the start. We did not isolate why start B has no divergences.

Sampling-free reference. MF and Jastrow converge within the L-BFGS iteration cap; all RBM runs stop at the cap (Table VIII), so the RBM row of Table I is an upper bound that is not tight: at 8/0.5 it is above the VMC median. In the other twelve tasks the RBM-SR median is above the L-BFGS value, by a factor of about 3 at 10/0.5 and about 400 at 10/0.25, so the RBM numbers in this paper measure the optimiser and the sample budget more than the ansatz.

SR shift and sample count. Two sweeps use RBM $\alpha=2$ SR at $\eta=0.1$ from start S, at 10/1, seeds 0–4; their base setting ($\epsilon=10^{-2}$, $M=256$) is the start-S main configuration. Table VII shows SR is unusable at this rate for shifts 10^{-4} and 10^{-3} (all five seeds diverged), mostly stable at 10^{-2} (1 of 5 diverged), and stable but less accurate at 10^{-1} and 1. We did not retune η per shift, so the sweep ties the start-S divergences to the shift–rate combination but does not show that a smaller rate at a small shift would fail. The median ϵ_E falls at every step from $M=32$ (5.6×10^{-4}) to $M=512$ (2.1×10^{-5}): the error of the trained RBM is tied to the sample budget, and differences between systems at the 10^{-5} level should be read relative to it.

Training curves. Fig. 2 shows the Monte Carlo energy error during training at 10/1 from start S (median of three seeds). With SR the RBM curves flatten early, whereas RBM-SGD is still decreasing at iteration 300.

Checks and cost. In Table VIII, no finite run violates the variational bound, no selected state has $\epsilon_E > 0.1$ (13 start-S runs do), and the median gap between MC and exact energy is 2.4×10^{-5} for RBM $\alpha=2$ SR.

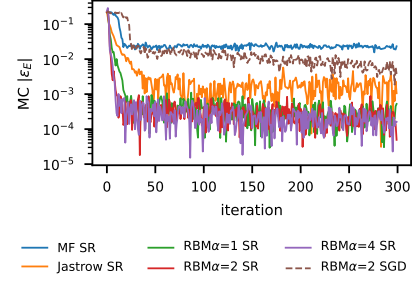


Fig. 2. MC energy error during training (10/1, start S).

VII. Limitations

Scale. One model, $N \leq 12$, 5 seeds, 300 iterations, 256 samples: nothing here extrapolates to large systems, 2D or frustrated models. Statistics. The paired tests are uncorrected and have little power. Post-hoc protocol. The two-start selection was introduced after registration and after an audit of single-start results; it uses the exactly evaluated energy, which a large- N study would not have. Tuning. Rates were tuned from start S only. Five of ten selected rates lie on a grid edge (MF with both optimisers and RBM $\alpha=2,4$ with SR at the top, Jastrow with SR at the bottom), so those systems might improve with a wider grid. SGD used no momentum or Adam (an adaptive optimiser could shrink the SR–SGD gap), and SR was not retuned jointly with the shift. Baselines. All are our own reimplementations, not NetKet or published code, without translation symmetry. For $\Gamma \leq 0.5$ the Jastrow VMC results remain above the Jastrow L-BFGS values (Table I), so that baseline is still under-optimised there. Reference. The L-BFGS reference uses one seed, covers MF, Jastrow and RBM $\alpha=2$ only, and did not converge for the RBM. Observables. Only $\langle |M_z| \rangle$ and fidelity at fixed N are reported ($\langle \sigma^x \rangle$ errors are logged but not tabulated), with no finite-size scaling or correlation functions. A full study would add NetKet-grade baselines, symmetric ansatzes, MinSR or Adam, larger N , more seeds, and tuning that covers both starts.

VIII. Conclusion

In a small VMC study of the periodic TFIM at $N = 8$ –12 with two starts per run, an $\alpha=2$ RBM trained by SR gave a lower mean energy error than Jastrow and mean-field states with SR, and than the same RBM with SGD, in all 13 tasks. The margin over Jastrow is large and backed by the sampling-free reference only for $\Gamma \geq 0.75$; for $\Gamma \leq 0.5$ it is below a factor of 2.5 and, in three of four tasks, reflects optimisation under this budget. At the critical field $\alpha=4$ was better than the method of record. The results depend on the start (Jastrow and mean-field fail from the symmetric start at small fields, and RBM-SR diverged only from it), on the SR shift and on the sample budget, and low-energy states in the ordered phase

TABLE VI

Start ablation: median ϵ_E over the non-diverged seeds of each start (S: symmetric, B: symmetry-broken; superscript: diverged seeds of 5); columns are N/T . SR systems only, for space.

System	start	10/0.25	10/0.5	10/0.75	10/1	10/1.25	10/1.5	10/2	8/0.5	8/1	8/1.5	12/0.5	12/1	12/1.5
MF SR	S	6.2e-5	1.5e-3	6.2e-3	2.3e-2	5.0e-2	6.8e-2	6.5e-2	1.1e-3	2.6e-2	6.7e-2	2.3e-1	2.2e-2	6.8e-2
	B	6.2e-5	1.3e-3	6.2e-3	2.3e-2	5.0e-2	6.8e-2	6.3e-2	1.1e-3	2.6e-2	6.8e-2	1.0e-3	2.2e-2	6.9e-2
Jas SR	S	4.0e-4	3.0e-3	5.3e-3	1.5e-3	1.9e-4	3.2e-5	2.6e-6	3.3e-3	1.2e-3	3.7e-5	2.8e-3	1.7e-3	3.0e-5
	B	5.6e-5	7.4e-5	8.4e-4	4.1e-3	1.9e-4	3.3e-5	2.6e-6	1.4e-4	1.6e-3	3.7e-5	6.0e-5	4.3e-3	3.1e-5
RBM $\alpha=1$ SR	S	6.1e-5	8.7e-5	1.2e-4	6.3e-5	1.3e-5	5.9e-6	7.2e-7	6.2e-5	3.3e-5	3.2e-6	8.8e-5	9.4e-5	6.4e-6
	B	7.3e-5	7.7e-5	8.0e-4	6.7e-5	2.0e-5	4.2e-6	7.2e-7	1.2e-4	3.7e-5	4.0e-6	6.0e-5	1.4e-4	5.9e-6
RBM $\alpha=2$ SR	S	5.2e-5	5.2e-5	8.1e-5 ¹	4.0e-5 ¹	5.9e-6 ¹	2.2e-6 ²	1.4e-6 ²	5.8e-5	1.7e-5	9.6e-7	6.1e-5	8.0e-5 ²	3.0e-6 ³
	B	6.7e-5	4.4e-5	7.6e-4	4.5e-5	6.1e-6	2.0e-6	1.7e-6	1.2e-4	2.7e-5	9.9e-7	3.7e-5	8.3e-5	3.2e-6
RBM $\alpha=4$ SR	S	2.0e-5	6.1e-5	9.7e-5	2.3e-5	5.4e-6 ²	2.5e-6 ³	2.1e-6 ⁴	5.2e-5	7.2e-6 ¹	1.1e-6 ²	5.6e-5 ¹	4.5e-5 ¹	6.1e-6 ²
	B	3.7e-5	4.1e-5	7.7e-4	2.7e-5	5.7e-6	2.3e-6	2.4e-6	1.2e-4	1.5e-5	1.1e-6	4.7e-5	6.5e-5	5.8e-6

TABLE VII

Sweeps of the SR shift ϵ (at $M=256$) and of the samples per iteration M (at $\epsilon=10^{-2}$), RBM $\alpha=2$ SR at 10/1 from start S: medians over non-diverged seeds; diverged seeds.

setting	ϵ_E	ϵ_{M_z}	$1-F$	div.
$\epsilon=0.0001$	–	–	–	5/5
$\epsilon=0.001$	–	–	–	5/5
$\epsilon=0.01$	4.0e-5	1.2e-3	9.1e-5	1/5
$\epsilon=0.1$	7.0e-5	1.2e-3	1.4e-4	0/5
$\epsilon=1$	3.8e-4	1.6e-3	7.4e-4	0/5
$M=32$	5.6e-4	2.5e-3	1.6e-3	1/5
$M=64$	2.6e-4	4.8e-3	7.4e-4	0/5
$M=128$	1.0e-4	3.0e-3	3.3e-4	2/5
$M=256$	4.0e-5	1.2e-3	9.1e-5	1/5
$M=512$	2.1e-5	3.9e-4	3.7e-5	0/5

TABLE VIII

Divergence counts, consistency checks, and run counts and times from the registry.

Quantity	value
diverged of 650 main runs, v1 / v2 (start S)	31 / 28
diverged of 650, final: start S / B / both	28 / 0 / 0
of these, RBM $\alpha=2$ / 4 SR, start S (of 65)	12 / 16
$\epsilon_E > 0.1$: start S / B / selected	13 / 0 / 0
min ϵ_E over the 1272 finite main runs	5.6e-7
median $ \epsilon_E^{MC} - \epsilon_E $, RBM $\alpha=2$ SR	2.4e-5
tuning runs per SGD / SR system	36 / 27
L-BFGS runs / stopped at iteration cap	78 / 26
reported / superseded or earlier runs	1773 / 1749
summed run time (min): reported / all	47 / 82
wall-clock with a run executing (min)	47
longest single run (s)	115

are often symmetry-broken. These findings hold for this testbed and tuning rule; they should be re-tested with an adaptive optimiser, per-start tuning, symmetric ansatz and larger systems.

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